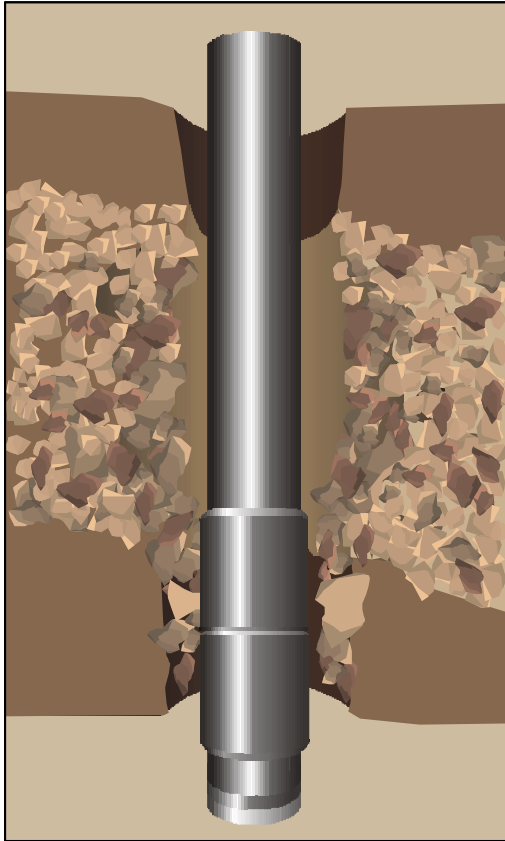




ABC of Hole Cleaning



NS-17

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1. Introduction

Analysis of stuck pipe problems by OCTG Procter Consultancy has shown that, in the majority of cases (60%), solids are the main sticking mechanism. As the number and complexity of long reach and highly deviated wells increases, this will stay the same, unless the appropriate steps are taken to ensure good hole cleaning practices.

Objectives

The objective of this book is to provide a basic knowledge of hole cleaning by providing an insight into what happens downhole, emphasizing the most common causes of poor hole cleaning and introducing an awareness of hole cleaning to a wider audience.

Why can Hole Cleaning be a problem

Just because solids are present in a well doesn't mean that stuckpipe problems are inevitable. After all, many wells have been drilled with much less consideration for holecleaning in the past than we currently give wells (probably because wells used to be restricted to 45 degrees or so).

As the angle of the well increases above 45°, the likelihood of a solids bed existing increases. However, the presence of the bed is usually only apparent once it starts to have an adverse effect on drilling, or when the drillstring is tripped out of the hole or pulled off the bottom for a check trip. Only then is remedial action usually taken. This action may be to alter the mud properties, circulate faster, rotate the drillpipe faster or make a wiper trip.

This has given rise to the following rules and procedures:

“If you're going to get stuck it will be in the first ten stands of a trips”

“If you pull into a problem while tripping out, go back down and circulate”

“The Tool Pusher must be on the drill floor while pulling out of open hole”

“The DSV must be on the drill floor for the first ten stands of a trip”

Often the existence of the solids bed is discovered too late. Evidence shows that the actions taken as soon as it is realised that the pipe cannot be retrieved often determines whether or not the drillstring becomes stuck.

The number of stuckpipe incidents will only decrease once the rig teams are aware of the signs of the solids bed and understand how to deal with it.

ABC of Hole Cleaning

The existence of a solids bed will become a problem when the rig team are not aware of its presence and misinterpret the feedback received while drilling. This leads to lack of action to prevent a more significant solids bed from forming, to remove the existing bed and the use of inappropriate methods to pull out of hole.

This manual aims to avoid stuckpipe incidents caused by solid beds by explaining the concept of hole cleaning in simple terms.

2. Life of a Cutting

Introduction

The aim of this section of the manual is to provide the reader with an understanding of the different type of solids, the difference between cavings and cuttings, slip velocity, settling velocity and the difference between Newtonian and Non-Newtonian fluids.

Creation of a Cutting

A cutting is a piece of rock which has been broken from the surrounding rock by the drillbit. The size and shape of the cutting depends on the type of drillbit, type of formation and the drilling parameters. The faster the drillbit cuts into the rock, the more cuttings are produced each minute.

Hole Cleaning

Hole cleaning is defined as the removal of solids from the well bore.

Solids

Solids are the debris present in the wellbore. It is made up mainly of the following:

Cuttings – rock cut away during the drilling operation.

Cavings – pieces of rock that have fallen away from the well bore.

Fines – a mixture of ground cavings and cuttings. Also known as Low Gravity Solids (LGS).

Swarf – pieces of metal cut away from casing or other metal present in the well.

Junk – anything in the well bore which should not be there

Cement – cement which has flowed into the wellbore and set.

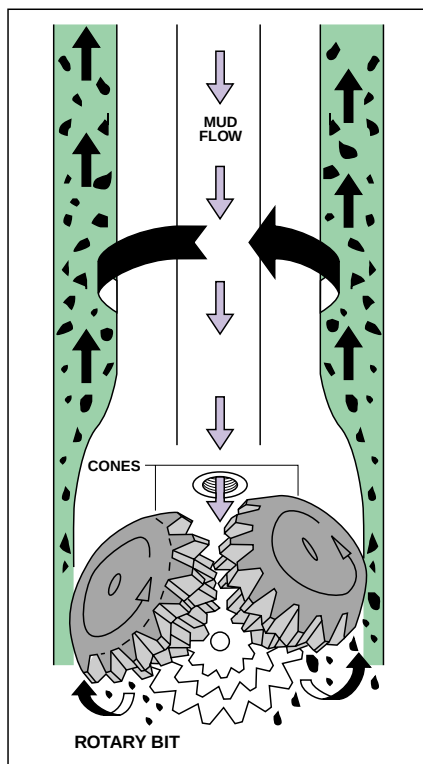


Fig.1 Cuttings are generated by the bit

Cuttings

Cuttings are transported out of the well relatively easily, and are the most significant solid in wells where no cavings exist. Cuttings can vary in size, depending on the drilling conditions. Very small cuttings, such as the ones which pass through the shaker screens, are called fines.



Fig. 2 - Sample of Cuttings

Fines

Fines, or Low Gravity Solids, represent the most significant contaminant of the drilling fluid system. These account for the major proportion of drilling fluid maintenance costs. The adverse effects caused by the fines include:

- Reduced ROP
- Problems with fluid rheology
- Increased wear in drilling components
- Increased risk of differential sticking
- Increased circulating pressure losses
- Increased time to remove fines by circulating



Fig. 3 Drilled fines in a tin - viewed from above

Fines are created by cuttings, which settle to the lower side of the wellbore. The action of the rotating drillstring crushes the solids and grinds them to form fines. The fines are usually seen on the lower screens of the shaker. Evidence of their presence will also be found when a retort test is performed by the mud engineer to calculate the LGS content.

Cavings

Cavings are pieces of rock which have fallen from the walls of the wellbore. They are generally much larger than cutting, typically 1" – 2", and are subsequently much more difficult to clean from the hole. The shape of a caving is typically flat or oblong, and considerably wider than they are thick. The following photograph shows typical shale cuttings.



Fig. 4 - Assorted cavings.

The following photograph shows smaller cavings, the size of which can be clearly seen.



Fig. 5 - Smaller cavings being held by an observer

Swarf

Swarf is small shavings of metals which are produced by milling operations or by unintentional metal-to-metal contact between drillstring components and casing. Large amounts of swarf are difficult to remove from the wellbore due to their weight and size. Swarf removal is not covered in this book.

Junk

Junk is any material that is unintentionally left in the well, such as dropped metal components or parts of drillstring components. Junk is not covered in this book.

Cement

Cement which has entered the wellbore and hardened behaves like cavings and can be dealt with in a similar manner. The problem of cement falling into the wellbore can be reduced by the use of fibre-based cement. Cement problems are not covered in this book.

Hole Cleaning

As this book is concentrating on the most frequent stuck pipe mechanism, it will deal primarily with the cleaning of cuttings and cavings from the hole.

Cuttings

The shape of a cutting may change during its journey from the drillbit to the shale shakers, for several reasons.

Clay cuttings may be affected by the drilling mud and may swell, causing the attractive forces between the clay particles to reduce, subsequently causing the cutting to break down. This breakdown process occurs when using a non-inhibitive water-based mud, such as Gyp-Ligno.

The breakdown of the clay may be so extreme that it becomes part of the mud, causing problems with the mud properties. This means that the mud must be diluted to maintain its performance. In fact, in wells where holecleaning is likely to be a major issue, the breakdown of the clay to become part of the mud can be advantageous, since the cuttings are easily removed from the hole. The degree of breakdown of the cuttings depends on the properties of the mud, in particular, its inhibitive properties.

ABC of Hole Cleaning

The action of the drillpipe and the bottom hole assembly will also break up the cuttings into smaller particles. The greater the time that the cuttings are downhole, the more they will be broken into smaller pieces.

Volume of Cuttings

The volume of cuttings generated is determined by the hole size and the rate of penetration (ROP).

As can be seen, larger drillbits generate a significantly greater volume of cuttings than smaller bits.

The following table illustrates the volume of cuttings produced in one hour. The volume of an average estate car is about 75 ft³

Hole size (in)	ROP (ft/hr)	No. of Cars filled per hour
17.5	200	4.45
17.5	100	2.23
17.5	50	1.11
12.25	200	2.18
12.25	100	1.08
12.25	50	0.54

Cavings

Cavings occur when the hole walls become unstable. When this occurs, large amounts of cavings can be generated in a short time. Generally, cavings are controlled by increasing the mud weight. The increase required varies for each case, but, in the early stages, an increase of 20 pptf of mudweight is usually sufficient. If the signs of cavings aren't detected early enough, and the hole condition is allowed to deteriorate, then much higher mud weights are required.

Transporting the Cutting to Surface

Once the action of the drillbit has broken the cutting away from the rock, it forms part of the solids in the well. Once it has left the bit area, the cutting is suspended in the drilling fluid and carried out of the wellbore with the fluid.

The cutting may be deposited and picked up several times before finally exiting the well. This process is dealt with in more detail in section 4, Solids

Transport. The remainder of this chapter will deal with the basic principles involved with transporting the solids to the surface in the drilling fluid.

Fluid Types

One of the many things investigated by Sir Isaac Newton was fluid flow. As a result of his experiments, Newton decided that there are two type so fluid: Newtonian and Non-Newtonian.

Newtonian fluids have a constant viscosity, regardless of the agitation given to the fluid. An example of this is water.

Non-Newtonian fluids exhibits the property that its viscosity changes as it is agitated. An example of this is Tomato Ketchup, which is thick and unpourable initially, but becomes thinner and pourable when the bottle is shaken.

Most drilling fluids are Non-Newtonian.

Solids in suspension

The term Solids in suspension means that the solids are only in contact with the suspension fluid (in this case the drilling mud) and not in contact with each other.

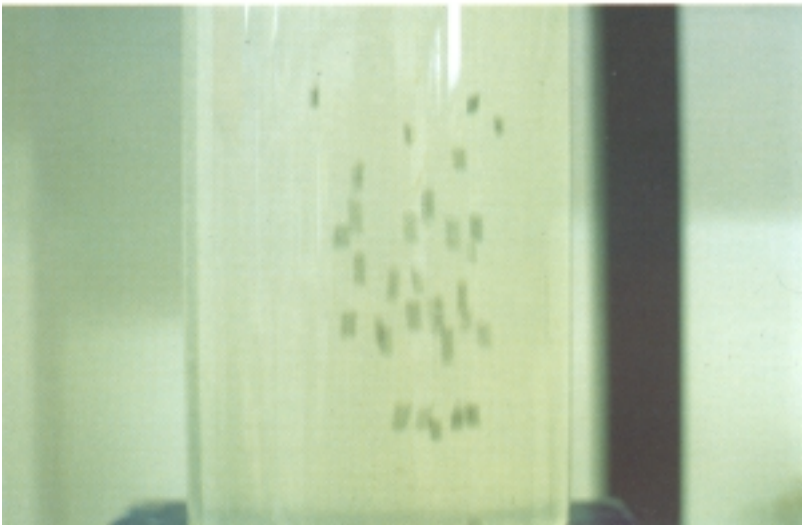


Fig. 8 - Particles falling through a stationary fluid

Due to the force of gravity, particles in the fluid will tend to move down through the fluid. If the fluid is stationary, the speed at which these particles move is called the Settling Velocity. If the fluid is in motion, the speed is called the Slip Velocity. In most drilling fluids, since they are non-Newtonian, the slip velocity is greater than the settling velocity. For Newtonian fluids, e.g. water, the slip velocity and settling velocity are equal. The slip velocity is constant for a given fluid and a certain density and size of cutting.

When the drilling fluid is stationary (the pumps are off), the cuttings fall vertically under the influence of gravity. When the fluid is moving (the pumps are on), the cutting will move out of the well, but more slowly than the fluid, since they are still slipping back, relative to the movement of the fluid. The direction of flow of the drilling mud is dependent on the angle of the well, but the direction of the slip for the solids is always vertical.

The fact that the solids are always slipping, both when the fluid is stationary and when it is flowing, is an important concept in understanding how downhole problems arise.



Fig. 9 - Particles falling through fluid in wellbore at an angle

The photograph above shows how the solids will fall vertically, under the influence of gravity, irrespective of the angle of the wellbore.

3. Roughnecks guide to Drilling Fluids

The aim of this section is to explain the functions and properties of a drilling fluid. It consists of two parts, Rheology and fluid flow

Roughnecks guide to Rheology

Rheology - The study of flow and deformation of matter

Drilling fluid has many characteristics. Listed below are those which are most relevant to hole cleaning.

Drilling Fluid Functions:

1. Cutting removal from the well bore.
2. Holding the cuttings in suspension while circulation is suspended.

Drilling Fluid Properties:

1. Weight.
2. Characteristics during flow.
3. Solids suspension ability.

The mud engineer will check the characteristics of the mud several times a day. The properties checked which are relevant to hole cleaning are:

1. Weight (Density)
2. Viscosity
3. Gel Strength
4. Methylene Blue Test (MBT)
5. Solids Content (HGS/LGS)
6. Sand Content

The following paragraphs discuss these properties.

1. Weight (Density)

This is discussed in section 5.

2. Viscosity

The ‘thickness’ or ‘runniness’ of the drilling mud is known as the viscosity. It is a measure of the resistance of the fluid to flow. It is also a measure of the ability of the fluid to carry cuttings.

Low viscosity = very thin - for example: Petrol

Medium Viscosity = Drilling Mud

High viscosity = very thick - for example: Syrup

Regular measurement of mud viscosity at the rig site is made using a Marsh Funnel Viscometer (Fig 10). This is a funnel-shaped device, sized such that two pints (US) of fresh water at 70 ± 5 °F takes 26 seconds to flow through it. Fluids that are more viscous will take longer, while those with a lower viscosity will pass through more quickly.

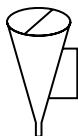


Fig. 10 - March Funnel Viscometer

When the mud engineer finds viscosity changes in the mud, a more detailed analysis will be performed to determine the cause of the change and whether any other characteristics of the mud have been affected. The engineer may also recommend action to restore the fluid's properties.

Rheology

As so often happens in the oil industry, the word Rheology has a more specific meaning than its dictionary definition. In the industry, the word Rheology is used to describe the 'thickness' or 'viscosity' of the drilling fluid at various flow rates. Viscosity of the fluid will change as the flow rate changes (the fluid is Non-Newtonian). This change is generally illustrated using a graph of viscosity against flow rate. Most drilling fluids have a graph similar to that shown below. Here, the viscosity reduces as the flow rate increases and is termed 'Shear Thinning' as it thins with increased shear.

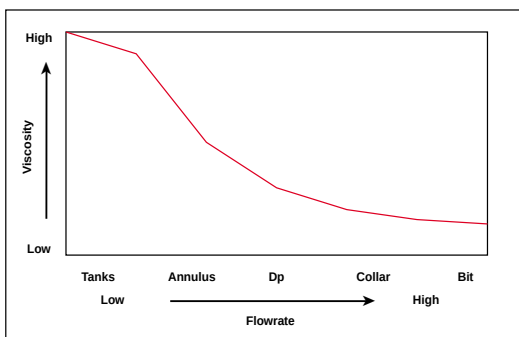


Fig. 11 - Typical Viscosity of Drilling Fluid from Tanks to Bit

The graph shows how the flow rate will vary, depending on the place where it is observed.

At the rig site, rheology measurements are made using a Fann Viscometer. This device provides a reading at various RPM. The readings are normally taken at settings of 600, 300, 200, 100, 6 and 3 RPM. The device simulates the fluid's flow properties under downhole shear rate conditions.

PV - Plastic Viscosity

Plastic Viscosity is the measure of the force required to maintain the flow of the drilling fluid once it has started to move. This simulates the mud flow in the drillpipe and at the bit nozzles (high shear areas).

The PV value is calculated as follows:

$$PV = 600 \text{ RPM Fann Viscometer Reading} - 300 \text{ RPM Fann Viscometer Reading.}$$

The PV is measured in centipose (cP).

The PV reading is proportional to the amount, size and shape of the solids in the mud. It indicates the size and number of fines in the drilling fluid. An increasing PV reading can be due to a buildup of solids in the mud, e.g. since the cuttings have not been cleaned from the well, they are ground into smaller fines.

YP - Yield Point

The Yield Point is a measure of the force required to start the fluid flowing from stationary. It is representative of the behaviour of mud in areas such as the annulus (low shear areas).

The YP value is calculated as follows:

$$YP = 300 \text{ RPM Fann Viscometer Reading} - PV$$

The YP is measured in lb/100 ft².

The YP reading indicates the chemical and physical attractive forces between the fines in the drilling fluid.

The apparent Viscosity is measured in centipose (cP), and is calculated by

$$\frac{600RPMF_{ann}}{2}$$

Gel Strengths

In simple terms, gel strength is an indication of the attractive forces between particles when the fluid is not flowing (static).

Initial Gel

This is the gel strength after ~0 seconds of rest.

TSG - Ten Second Gel

TSG is a measure of the attractive force present in the drilling fluid after it has been stirred for 30 seconds in a Fann viscometer at high speed, then left undisturbed for 10 seconds. The maximum reading obtained after switching on the viscometer is the TSG. It is the force required to re-initiate movement in the fluid after 10 seconds of rest. The reading indicates how well the mud holds cuttings in suspension.

TMG - Ten Minute Gel

TMG is the reading from the viscometer once the fluid has been stirred for 30 seconds at high speed, then left undisturbed for 10 minutes. The maximum reading obtained after switching on the viscometer is the TMG. It is a measure of the force required to restart circulation and to restart movement in the drilling fluid after circulation has ceased for 10 minutes. The reading provides an indication of how difficult it is to break circulation.

Impact of Gels on Hole Cleaning

The ideal drilling fluid is one that will remove all drilled cuttings in one circulation. However, as the well bore increases in length or difficulty, the chance of well cleaning problems will increase.

Based on experience from North Sea extended reach wells, it has been discovered that hole cleaning is aided by raised low end rheology (i.e. high gels). These prevent the settling of cuttings into a cuttings bed and reduce the risk of avalanching.

There are several types of gel, each with different characteristics:

Gel Type	Characteristics	TSG/TMG
Fragile Gels	Poor Suspension	2/3
Good Gels	Good Suspension	5/9, 6/11
Progressive Gels	High Swab and surge pressure indicates buildup of solids	6/35, 15/60
Flat Gels (flash gels)	Good suspension but indicates flocculation	14/15, 23/25

Observation of the trends of the rheological properties of the drilling fluid is important, as they will provide indications of any hole cleaning problems that may occur.

MBT - Methylene Blue Test.

This test is performed to quantify the amount of reactive clay in water based drilling muds. A high level of clay in the mud may indicate potential problems with the formation being drilled.

Clay balls and shaker blinding may occur when drilling a reactive formation. The initial reaction to this is often to reduce the flow rate, preventing the shakers from blinding and losing mud. However, if currently circulating bottoms up, this is not a good idea, since the BHA may start packing-off with solids falling onto it.

MBT should be less than 30 lbs/bbl benonite equivalent. MBT is also called CEC - cation exchange capacity.

HGS/LGS Content

LGS - Low Gravity Solid, drilled cuttings with an average weight density of 2.6 kg/l

HGS - High Gravity solids, weighting agents like barytes with a density of 4.2 kg/l

In most drilling operations, the level of low gravity solids in OBM should be less than 10%. For WBM, the LGS should be less than 6%. A higher level of LGS is acceptable when an inhibitive WBM is in use, due to the high cost of maintaining it at 6% LGS.

Sand Content

It is important to keep the sand content of the drilling mud as low as possible, ideally below 1%. High sand content is a contributory factor in equipment failure due to erosion, as well as causing mud related problems.

Roughnecks guide to fluid flow

Fluid flow can be described as one of two types: Laminar, which is smooth and slow; or Turbulent, which is fast and erratic. To illustrate the difference between these, think of a river:

Laminar Flow: When the river flows through a wide valley, the flow is smooth and slow with few ripples on the water. Some of the gravel and grit picked up by the river is deposited on the river bed and at the river banks. Laminar flow has a lower pressure loss than turbulent.



Fig. 12a - Laminar Flow

Turbulent Flow: When the river flows through a narrow gorge, the flow is more disturbed and turbulent. Rocks and gravel picked up by the river will be held in the flow to be deposited downstream in a slower flowing laminar region.



Fig. 12b - Turbulent Flow

The effect of viscosity on turbulence

The ‘Reynolds Number’ of a fluid is a measure of how difficult it is to make the fluid go into turbulent flow. With higher viscosity liquids (i.e. more syrupy ones), it is more difficult to make the flow turbulent.

Newtonian Fluids

Newtonian fluids are the simplest fluid type, such as water. In such a fluid, the shear stress (the measure of how difficult as fluid is to stir) is directly proportional to the shear rate, while the flow is laminar. In other words, to make the fluid flow twice as fast, you need twice as much energy.

A Newtonian fluid will start to move as soon as pressure or force is applied.

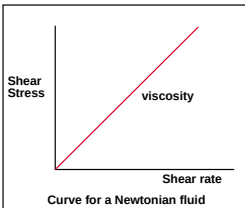


Fig. 13a Newtonian Fluid

Non-Newtonian Fluids

Most drilling fluids are Non-Newtonian fluids. They contain solids which form a gel structure between the particles. As the shear rate increases, the shear stress increases until the resistance to flow is overcome. This point is known as the Yield Point. In other words, it's more difficult to start the fluid moving than it is to keep it moving.

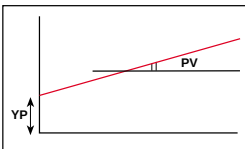


Fig. 13b - Non-Newtonian Fluid

A Non-Newtonian fluid requires initial pressure or force before it starts moving.

ABC of Hole Cleaning

The following tables illustrate how the variations in the different mud properties affect the performance of the various functions of the mud.

		FUNCTIONS				
PROPERTIES	Affect of properties on functions	Solids Removal	Solids suspension	Hydraulics	Lubrication	Hole Stability (Shale)
	Density	Y	Y	Y	Y	Y
	Viscosity		Y	Y	•	N
	Gel strengths	Y	Y	N	N	N

Y = Does affect N = Does not affect • = Some affect

Fig. 14a - Table showing whether the property affects the function of the drilling mud

		FUNCTIONS				
PROPERTIES	Affect of properties on functions	Solids Removal	Solids suspension	Hydraulics	Lubrication	Hole Stability (Shale)
	Density ↑	↑	↑	↓	↓	↑
	Viscosity ↑	↑↓	↑	↓	•	X
	Gel strengths ↑	↑	↑	X	X	X

↑ = Up or Better ↓ = Down or Worse ↑↓ = Complex relationship • = Some affect

Fig. 14b - Table showing the relationship between property and function when the property is altered

4. Solids Transport

The aim of this chapter is to describe to the reader the characteristics of cutting beds, the effect known as the Boycott effect and its influence on drilling and avalanching, the characteristics of cuttings beds at various hole angles and the effect of rotation and reciprocation on them and the value of annular velocity in hole cleaning.

Solids transport can be defined as the movement of cuttings and cavings out of the well bore. The manner of transportation depends on the angle of the well, the mud rheology and the fluid flow characteristics.

Vertical Wells

In the case of a vertical well, the flow is straight up. As can be seen from the photo below, the solids fall in the opposite direction to the direction of flow of the fluids. As long as the fluid is flowing up the well at a faster rate than slip velocity, the solids will be cleaned out of the hole. If the flow is stopped, the solids will begin to fall back down the well (settling velocity). If the fluid is stopped for long enough, then the solids will reach the bottom of the well and begin to build up. This buildup gives rise to the term “fill”. For example, 10 ft of fill means that the bottom 10 ft of the well has filled up with settled solids.



Fig.15 - Cuttings transport in a vertical well

The time taken for the solids to settle to the bottom of the well depends on the gel strength of the drilling fluid. It usually takes quite a long time for the cuttings to reach the bottom of the well, since the distances involved are quite large (several hundred feet).

Highly Deviated Wells

Since solids always fall vertically under the influence of gravity, in a highly deviated well, they have considerably less distance to fall, usually only several inches to the low side of the well when there is no fluid flow (pumps off). The actual time for the solids to reach the borehole wall depends on the gel strength of the drilling fluid, but is obviously a lot less than a vertical well. The layer of solids on the low side of a deviated well is known as the cuttings bed. This effect is called the Boycott effect and was discovered by a Doctor Boycott while observing the separation of red and white blood cells. He noticed that if the test tubes containing the blood were angled, the separation occurred more quickly.

This explains why hole cleaning is so important in deviated wells. Currently, the main reason for BHA becoming stuck in deviated wells is solids related.



Fig. 16 - Cuttings settling in a deviated well

Summary

In all wells, the solids fall vertically down as the fluid moves up. It is easy to see how, in a vertical well, the solids will take a longer time to build up than in a deviated well. From the two descriptions above, it is possible to visualise how the solids will build up in a well of any deviation from vertical to horizontal. The closer that the well becomes to horizontal, the quicker the cuttings bed will build up.

Solids or Cuttings Beds

The term solids or cuttings bed is used throughout the industry to mean beds made up of all types of solids (i.e. cavings, cuttings etc).



Formation of Cutting Beds

A cuttings bed will form when solids drop to the low side of the well and the flow rate of the drilling fluid is insufficient to pick them back up into the flow and hold them in suspension.

As the angle of the well increases above 35° , the cuttings beds will be more significant. At angles between 25° and 65° , the solids in the bed are more loosely packed, and more likely to avalanche down the well.

Avalanching

Cuttings bed avalanching occurs in a similar way to snow avalanching. The effect can be visualised by thinking of snow. Snowflakes fall fairly slowly at about 10 MPH. It settles on the hillside with millions of other flakes. When an avalanche occurs, tons of loosely packed snow rolls down the hillside at around 70 MPH. These are the same snowflakes that fell through the same air at 10 MPH. Why the difference ? The secret is that the air inside the avalanche is moving and forms the mass of its volume. The only friction is at the surface of the moving snow.

Similar conditions exist in a cutting bed avalanche. The problem can occur when the flowrate is high or when it is zero. Avalanching is most likely to occur in wells with angles of between 45° and 65° . At angles above this, the cuttings bed is generally stable. At angles below 45° , avalanching may still occur if circulation is insufficient to clean out the solids. Increasing the low end rheology can reduce the tendency of the cuttings beds to avalanche.

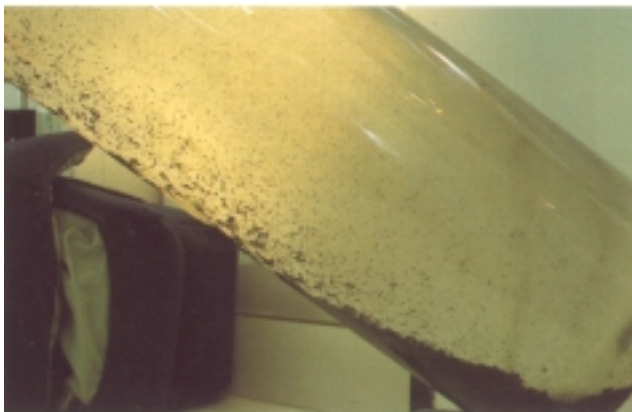


Fig.17 - Cuttings Bed avalanching in a deviated well

Stable Cuttings Beds

At angles between 65° and 90° , any formed cutting beds will be stable. A stable bed often cannot be removed by increasing the flow rate of the drilling fluid, and some sort of mechanical action must be used. This can be the action of rotating the drillpipe or removing the drillstring from the hole. Whether back reaming is in progress at the time or not, the action of pulling the BHA will either agitate the cuttings bed, or drag it higher up the well.

A check trip will stir up the cuttings bed. However, this may have a good or bad effect. If the bed is agitated, it can then be circulated out, using the appropriate techniques. If the solids are not circulated out, there is a danger that the cuttings will move up the well and form a new cuttings bed in conjunction with an existing bed. It is also possible that the cuttings will fall back onto the BHA during the next circulation stop.

The formation of cuttings beds can be minimised by the use of appropriate methods while drilling and circulating solids out of the well.

Height of Cuttings Bed

The height of the cuttings bed may cause problems when drilling, but what height is too high?

The answer is dependant on the size of the BHA. It can be best visualised by considering the percentage of the annulus which is filled by the solids. A 10% cuttings bed can cause severe problems when drilling any size of hole.

The thickest cuttings beds are usually found in well with angles of between 45° – 65° . At this point, solids coming out of the hole are meeting other solids which are avalanching down the hole.

The existence of a 60% cuttings bed in a 17.5" hole (i.e. 10 inches high) has been found in a 55° section.

Effect of String rotation and reciprocation

Rotation of the drillstring will stir up the cuttings beds. The amount of agitation depends on the rotational speed, but evidence shows that there is a dramatic difference between no rotation at all (i.e. sliding) and some rotation (i.e. 40 – 60 rpm). Results show that increasing the rotational RPM increases the effectiveness of the hole cleaning, but it is not known if this is a linear relationship. It may be useful to attempt hole cleaning at different speeds for a particular well and to use the results to optimise further operations on the well.

For rotation & reciprocation to be effective, they must be used in conjunction with other hole cleaning techniques such as circulating high density/low density pills and circulating for several fluid bottoms up times as well as the use of an appropriate flowrate.

Large Holes

Problems may occur on floating drilling units with cuttings building up in the riser due to its large diameter. This causes the annulus loading to become too high and losses occur. It is recommended that a booster line be used where high levels of solids loading in the riser and BOP is expected.

ABC of Hole Cleaning

The size of the casing, BOP and riser all effect hole enlargement and annular velocity. Since these are fixed, the rig crew have little influence over their size. Other factors, such as pilot holes and washouts are variable and may be uncontrollable.

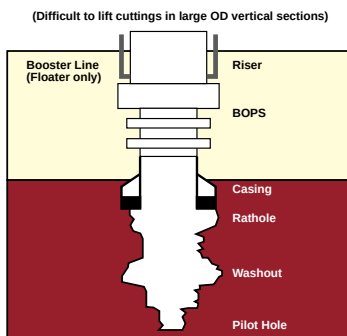


Fig.18 - Type and locations of washouts

To illustrate the influence that washouts can have:

If washout increases the hole size from 17-1/4" to 20", the volume of rock increases by 166% and the flow rate drops by 281%.

Washouts

Cuttings may get trapped in oversize areas, known as washouts, on their way to the surface. In these enlarged areas, the velocity of the drilling fluid slows. This may cause the slip velocity to become greater than the fluid velocity, and the cutting will settle in the washout area. These cuttings can build up until

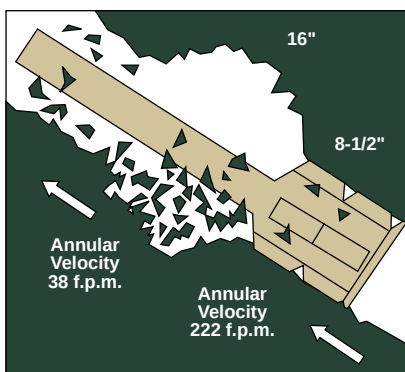


Fig. 19 - Illustration of washouts effecting hole cleaning.

they fall back into the fluid path and appear at the surface as slugs of cuttings (intermittent or erratic returns).

Annular Velocity

Annular velocity is defined as the speed of the drilling fluid in the area between the drillstring and the casing or wellbore (the annulus).

It can be calculated using the following formula:

$$AV = \frac{24.51 * GPM}{(Holesize^2 - DPsize^2)} ft/min$$

The effect can be visualised by comparing it with a river running through a wide valley at a rate of, say, 10000 gallons per minute. Where the valley is wide, the river flows more slowly. However, when the valley narrows, and the river flows through a narrow gorge, the flow rate remains the same, but the speed must increase, since the same amount of water has to flow through the narrow gap. This gives rise to turbulent flow, seen as the presence of rapids.

This can be applied to the wellbore. The annulus between the BHA and the wellbore is the gorge, where the speed of flow is high. The annulus between the drillpipe and the wellbore is the wide valley, where the speed of flow is lower, and the washed out sections of the bore are similar to lakes where the speed is very slow.

	AV (ft/min)	Flowrate (GPM)	Hole (in)	DP (in)
(a)	61.00	700	17.5	5
(b)	78.43	900	17.5	5
(c)	97.99	500	12.25	5
(d)	102.28	500	12.25	5.5
(e)	146.99	750	12.25	5
(f)	156.79	800	12.25	5
(g)	207.49	400	8.5	5

The table above shows annular velocities for various drillpipe and hole sizes. It is recommended that the annular velocity is not allowed to fall below 150 ft/min.

Effect of Hole Angle on Annular Velocity

In a vertical well AV1 is equal to AV2* for a given flow rate Q . However, in a deviated well with a cutting bed, AV3 is higher than AV1 or AV2. The fluid takes the path of least resistance, in this case the larger area above the drillpipe.

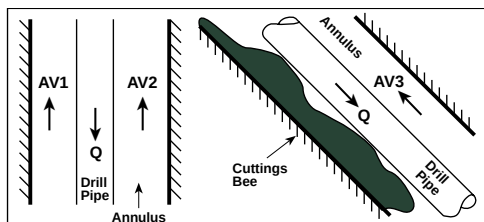


Fig.20 - Effect of hole angle on annular velocity

*Where the drillpipe is close to one side of the well bore, AV1 and AV2 may be different.

The illustration below shows the flow rates at the different points in a cross section of the wellbore, with the drillpipe lying on the low side of the hole. The fluid will tend to stagnate in areas A & B, causing the solids to fall out of the fluid more than when the drillpipe is centred in the hole. The action of rotating the drillpipe will cause the solids to be stirred up into the higher flow areas and transported out of the well.

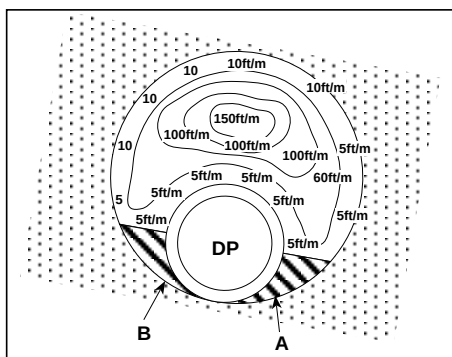


Fig. 21 Annular velocity profile

5. Hole cleaning

The objective of this section is to provide an overview of the impact that the various field-controllable parameters have on hole cleaning, along with specific hole cleaning problems which may occur at various hole sizes.

The previous sections have provided information on the various mechanisms involved in hole cleaning. This section covers the main topic of hole cleaning.



Fig.22 - View down the cuttings bed sticking model

The picture above shows the scale model used as part of the Stuckpipe Training Course. It represents a 1/3 scale model of a 12.25" hole and is made of clear Perspex. Salt is used to simulate the solids – sea salt (2 – 4 mm) for solids of 6 – 12 mm and table salt (0.5 – 1 mm) for solids of 1.5 – 3 mm. The cuttings represent a bed of 5 – 10% (1.5"). This is a conservative estimate for a deviated well, where beds of up to 4" could be expected.

The chart below illustrates how each of the field controllable parameters influences the hole cleaning.

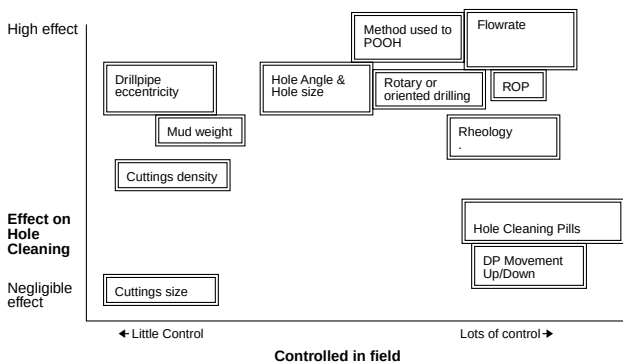


Fig.23

From the chart, it's possible to produce a list of the parameters which are critical to good hole cleaning. The list below is ordered to put the items which have the greatest effect and which the rig team have the greatest control over first.

- a. Annular Velocity (Flowrate)
- b. Drillpipe Movement
- c. Rheology
- d. Hole Cleaning Pills
- e. Hole Angle
- f. Methods used to POOH
- g. Mud weight.

This section will consider each of these in turn.

The Effect of Flow Rate

Flow rate is by far one of the most significant controls that the rig team have over hole cleaning. Since a cuttings bed is difficult to remove once formed, the best practice is to stop it forming in the first place. This can be achieved using a high flow rate, optimum rheology and correct drillpipe movement.

The maximum flow rate is restricted by the hole size, drillpipe size and the maximum surface pressure. It is controlled by the driller, who should always aim to maximise the flow rate, unless there are conditions which override the importance of hole cleaning. This is especially important in the large hole sizes, where even the maximum flow rate may not be sufficient to clean the wellbore.

If the flow rate is reduced for any significant length of time, circulation may need to be started bottoms up again, since the solids which were dispersed throughout the well before circulation was stopped will have settled to the bottom. Experiments have shown that cuttings slurry moves out the well slower than the fluid velocity, up to 3 – 5 times slower in the case of a deviated well.

Occasions may arise where it is necessary to reduce the flow rate for operational reasons. In these cases, all attempts should be made to maintain the maximum obtainable circulation rates.

Effect of ROP

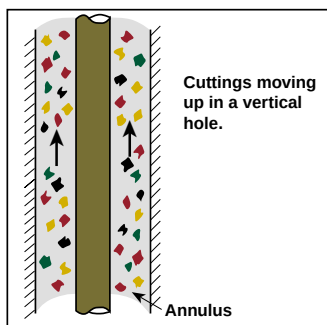
The rate of penetration must be closely controlled to prevent the volume of cuttings generated becoming so high that they drop out of the drilling fluid at a high rate. A cuttings volume of 4% in a vertical well, reducing to 0.5% in a 60° well is desirable.

Rheology

The two main properties of the drilling fluid which provide optimum drilling performance are viscosity and gel strength, as these are directly related to cuttings suspension and transport.

Vertical and low angle wells

In wells with angles of between 0 and 30°, hole cleaning is directly related to flow rate. As the angle increases, the hole becomes more difficult to clean.



At low angles, the hole can be cleaned without any special requirements. The drilling fluid is required to carry the cuttings out of the hole and keep them in suspension until the pumps are stopped.

Where poor hole cleaning is detected, usually by a build up of cuttings at the bottom of the well, a viscous pill can be used to remove cuttings.

It is important to remember that if a pump failure occurs while pumping a pill, start circulating bottoms up from the beginning.

To clean a large well bore (i.e. 26" or 17-1/2") a high pump output with a good mud carrying capacity is required. For this reason, maintain as low a PV as possible to enhance the pump output.

Intermediate angle wells – 30° to 60°

The most difficult wells to clean are those with an angle of between 30 and 60 degrees. Because of this, it is important to try to prevent the cuttings beds from forming in the first place. One method is to run the high end mud rheology as low as possible, but still at a sufficient level to clean the vertical section. This will give the greatest turbulent action and will circulate the majority of the cuttings out of the hole.

A reduction in the rate of penetration is another option, as this reduces the level of solids loading in the wellbore. Experience has shown that rotation drilling is preferred to oriented drilling, as the mechanical action of the drillpipe increases the hole cleaning.

Turbulent flow increases the effectiveness of the hole cleaning. However, with large hole sizes it is often difficult or impossible to achieve this. A high low-end rheology is still required to prevent a cuttings bed forming when the pumps are off.

Reynolds Number

$$\text{Re} = \frac{\rho \cdot V \cdot \Delta d}{\eta}$$

ρ = density, V = annular velocity, Δd = diameter & η = viscosity

For a given well, ρ and Δd cannot generally be changed.

To keep a high Re number (turbulent flow), V should be large and η should be small. Therefore, ‘thin and fast’ is the preferred option as it increases the chance of the mud being in turbulent flow.

If turbulent flow cannot be achieved in the wellbore, then the cuttings must be removed using laminar flow. This is more difficult than with turbulent flow, and the rheology of the fluid becomes more important.

With well angles between 40 & 60 degrees, the cuttings bed will avalanche. This may occur when the pumps are on or off.

It is important to be aware that wells with a high inclination (i.e. 90°) also have an area where the angle is 40 – 60 degrees. This can be a problem area. It is also likely that this area is in the casing and the annulus size will be larger. Do not think that all problems are over once the BHA is in the casing.

Since the fluid takes the path of least resistance, in cases where the drillpipe is lying on the low side of the annulus, the fluid flow will be concentrated on the high side of the annulus. This means that the scouring action of the fluid on the cuttings bed will be dramatically reduced.

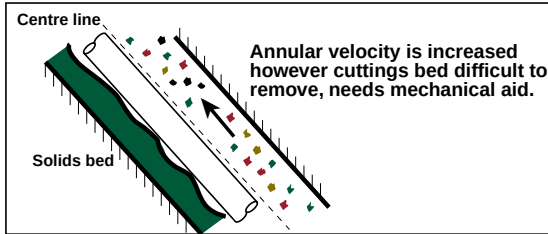


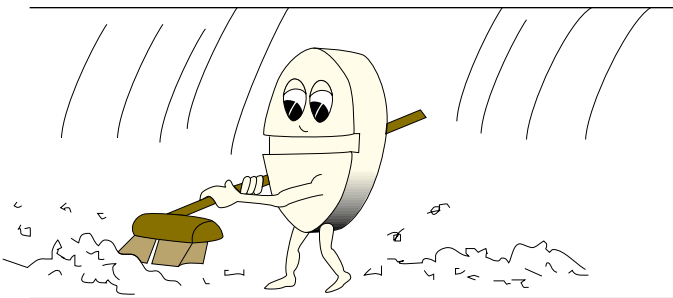
Fig.25

High Angle Wells – 60° - 90°

It is important to balance the mud's properties when drilling high angle sections. This often means that the final properties are a compromise. For example, a higher viscosity is need to transport the cuttings out from the vertical section, whereas a lower viscosity is required to stir up the cuttings in the high angle section.

Hole Cleaning Pills

There are two main types of cleaning pills – viscous pills and combination pills.

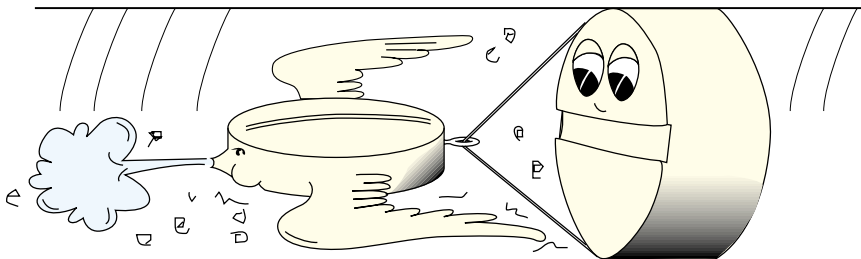


Viscous pills are generally used when drilling top hole and straight low angle sections. When used with water based muds, they are made from Guar Gum, XC polymer or bentonite, and can be weighted or unweighted, depending on the drilling fluid in use. A standard high viscous bentonite pill is still in use to sweep the hole of any residual cuttings.

ABC of Hole Cleaning

Combination pills are used in the highly deviated sections. These pills consist of a low viscosity brine, water based mud, base oil, oil based mud or pseudo oil based mud followed by a weighted viscous pill. The concept is to pump a balanced combination pill with equal density to the mud weight, and to follow it with a weighted pill which has a density of at least 100 pptf above the mud weight.

A viscous pill is used in deviated wells (up to 40 degrees), as in high angle wells, it tends to deform over the surface of a cuttings bed, rather than stirring it up.



A combination pill works as follows: First, the light weight pill causes turbulence which stirs up the solids from the low side of the hole. Then, the heavy weight pill sweeps the cuttings out of the hole.

It is important to make sure that the pills don't affect the overbalance on the formation, which may cause the well to flow.

POOH Methods

The concept of a check trip or tripping in a deviated hole is to 'check' that the hole is clean and to take action if it is not. Often, the trip is seen as the action itself.

Tripping or performing a check trip is best done by pulling the string out of the hole with the pumps switched off and with no rotation. The 30k overpull rule should be applied (see below). This method will allow the driller to obtain a good observation of the condition of the well.

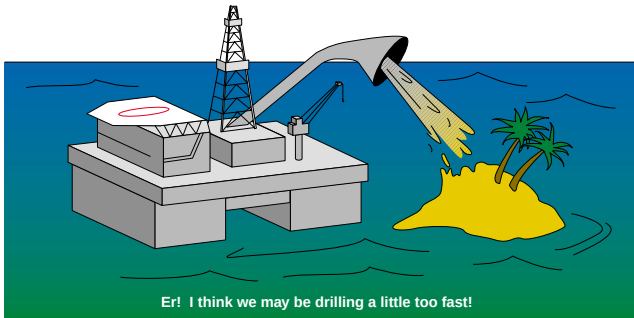
It has been observed that there is a relationship between a clean hole and low torque and drag figures. Torque and drag charts provide a good mechanism for observing these trends.

The 30k Overpull Rule

When pulling out of a well with an angle of greater than 35 degrees, the initial overpull should be limited to 30k lbs or $\frac{1}{2}$ of the BHA weight in mud, whichever is less. If 30k is reached, then the string should be moved up a short distance (1 stand or a single) and circulation bottoms up should be performed. If there is a problem which is not cuttings/solids related it may require alternative action.

Back Reaming

Should the problems become so severe that backreaming is the only solution for getting the string out of the hole, then the following should be considered.



Consider a deviated well with a 10% by volume cuttings bed.

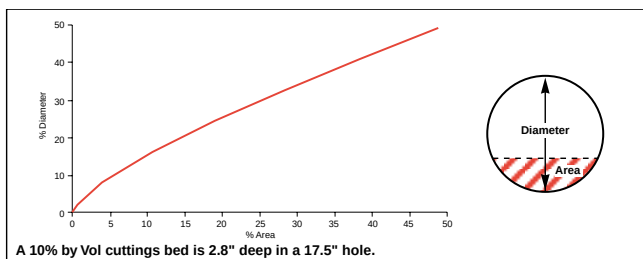


Fig.26 - Relationship between % Area & %Diameter for a circle

ROP is often limited when drilling to prevent the annulus cuttings from reaching a high concentration. The concentration of cuttings in the annulus cannot be easily determined and the figure generally used is the % in the mud at the bit. The % of cuttings on the annulus that is acceptable is dependant on the risk taken.

ABC of Hole Cleaning

Experience has shown that it is possible to drill wells in record time without problems using lower than recommended flow rates. This is considered high risk, until the process is fully understood.

While backreaming, we are effectively drilling out of the hole by stirring up the solids bed. The following example looks at the speed of operation and how the annulus may become overloaded while back reaming.

For example, consider a 17.5" hole with a deviation of $>40^\circ$. The maximum ROP for that section would be about 50 ft/hr. When backreaming through a 10% (volume) cuttings bed, the volume of solids stirred up is 10% of the volume produced while drilling.

So as to limit the concentration of cuttings in the annulus to the same level as when drillings, we can only backream at 10 times the rate we drilled (i.e. 10% times 10 = 100%, the volume initially drilled).

The following formula can be used to calculate the maximum backreaming rate (RBR) for any cuttings be size, where the maximum ROP for good hole cleaning is known.

$$\text{Rate of Backream} = \frac{100\%}{\%VolCuttingsBed} * \text{MaxROP}$$

For the example above of 10% cuttings bed and 50 ft/hr ROP:

$$100\%/10\% * 50 \text{ ft/hr} = 500 \text{ ft/hr} = 5.5 \text{ stands per hour.}$$

If a 20% cuttings be is present, and the maximum ROP is 50 ft/hr:

$$\begin{aligned} \text{RBR} &= 100\%/20\% * 50 \text{ ft/hr} \\ &= 5 * 50 \text{ ft/hr} \\ &= 250 \text{ ft/h} \\ &= 2 \frac{2}{3} \text{ stands/hr} \end{aligned}$$

In high angle wells, the largest cuttings beds are generally found in the 55° section. These have been found with depths of up to 10" in a 17.5" hole. Using these values and a maximum ROP of 80 ft/hr, the RBR becomes:

$$\begin{aligned} \text{RBR} &= 100\%/60\% * 80 \text{ ft/hr} \\ &= 1.67 * 80 \text{ ft/hr} \\ &= 133 \frac{1}{3} \text{ ft/hr} \\ &= 1.5 \text{ stands/hr} \end{aligned}$$

Hole Angle

As the chart below shows, the most difficult holes to clean are those with an angle of 55° . This is due to the formation of unstable beds at angles of less than 55° that avalanche down and settle out at higher inclinations.

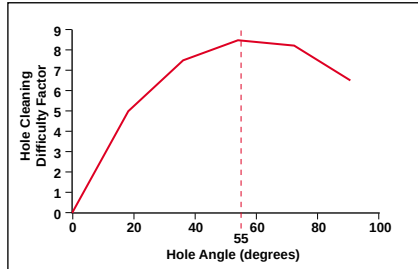


Fig.27 - Difficulty of Hole Cleaning with Hole Angle

Horizontal Wells

When cuttings are lifted by the drilling fluid at point A, they travel with the mud flow until the circulation stops, when it is deposited on the low side of the hole at B. It remains stationary until circulation starts again.

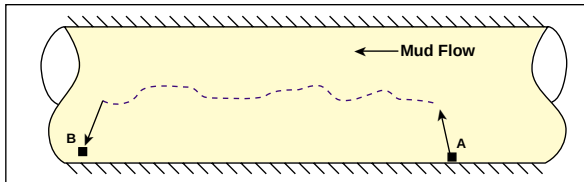


Fig.28a Cuttings lifted and deposited in a 65° - 90° Well

Deviated Wells

In the 55° section of the well, the cutting is picked up by the fluid at point A and deposited back at point B when the circulation stops. The cutting then falls back down the well to point C (often avalanching with other cuttings). The speed that the cutting falls from point B to point C is much faster than its slip velocity in the fluid.

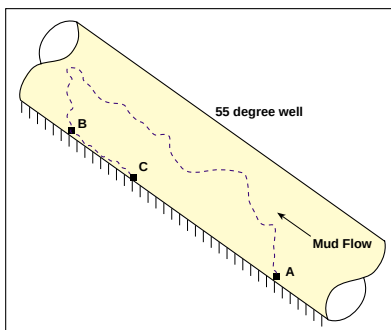


Fig.28b - Cutting Path in a 55° Well

Vertical Well

In this case, the cutting is carried from point A to point B, when the circulation is stopped. At this point, the cutting drops back to the bottom of the well. The rate of descent is the slip velocity.

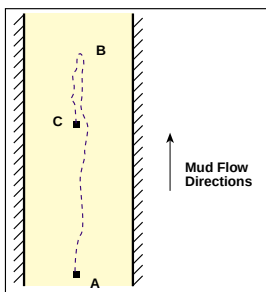


Fig.28c - Cutting Path in a Vertical Well

Drillpipe Movement

The rotary action of the drillpipe agitates the mud in such a way that it moves up the well in a spiralling manner.

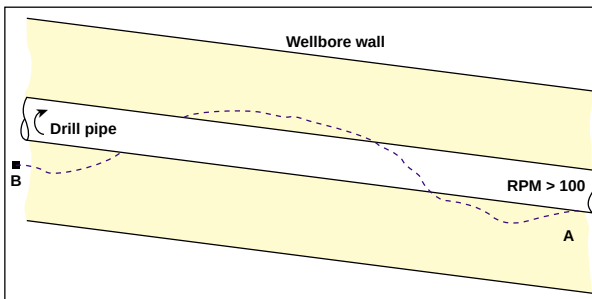


Fig.28d - Effect of Drillpipe Movement on cuttings

When the rotational speed of the drillpipe is low or it has stopped, the cuttings move to the low side of the hole under the influence of gravity. Because of this, cuttings beds will build up faster when drillpipe rotation is not used. It is noticeable that when drilling in oriented mode, it becomes difficult to get weight onto the bit due to the buildup of cuttings beds. This illustrates why drillpipe rotation is an essential aid to hole cleaning. Reciprocation of the drillpipe also helps hole cleaning by causing surges in the annular velocity.

However, it is important to be aware that reciprocation in unstable shales may cause wellbore instability.

Mud Weight

The mud weight provides an additional benefit when cleaning the hole, as a higher weight gives a higher buoyant force, which improves the carrying capacity of the mud. Increasing the buoyancy also slightly increases the ability of the fluid to lift cuttings from the low side of the well.

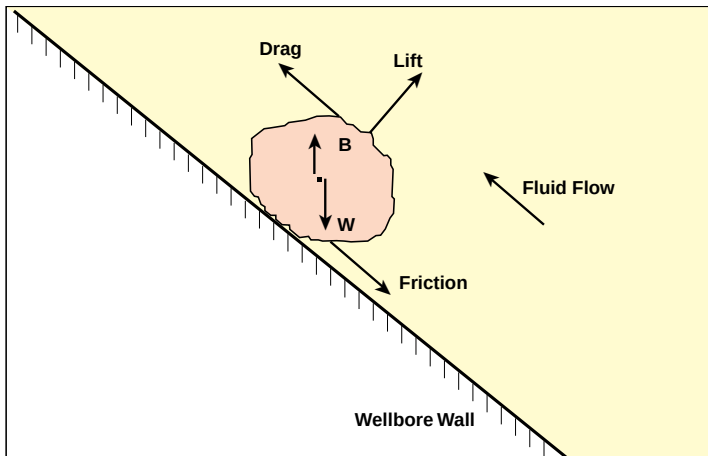


Fig.29 - Forces on a cutting

B = Buoyancy, D = Drag from fluid, F=Friction

W = Weight, L = Lift from fluid (aerofoil effect)

6. How a Cuttings Bed Acts while POOH

The section illustrates what happens to a cuttings bed as the BHA is pulled through it. It also provides information on the actions to be taken if sticking occurs when POOH.

A cuttings bed sticking model is used to illustrate what happens downhole while POOH in the presence of a cutting bed.

The Model

The wellbore is made up of two 2m sections of 100 mm diameter, 4 mm wall thickness Perspex tubing. The two tubes are joined using a 1m piece of the same tubing. The sharp edges are covered with tape.



Fig.30 - The model lying across tables

In the above picture, the model can be seen lying across three tables. The BHA is inside the tubing and the cuttings (salt) can be seen. The length of rope visible is used to pull the BHA through the tubing.

The stabilisers and bit have been machined to approximately a scale 1/8" undersize.

The model is used in a horizontal position to simplify the operation. This is representative of the best case for hole cleaning, as no avalanching will occur.

The model is operated with no fluid or fluid flow. A fluidised bed would flow more readily than a dry bed. Obviously, a dry bed is not the case in reality, but, although the distances, forces and times may vary, the mechanics of the operation do not change greatly. The model is sufficient for illustrating the basic principles of what happens downhole when pulling out without backreaming or circulating.

The BHA is made of six parts fabricated from Nylon and machined to 1/3 scale. All the components screw together with a common thread.

The thinnest section of the model is the 30 ft section of 5" drillpipe. Next to this is a 32 ft section of 8" drill collar. An alternative of 35 ft of 9.5" drill collar is also available.

The top stabiliser has 12.125" (slightly undergauge) blades on a 10 ft long, 9.5" body and is a straight bladed type not normally used in 12.25" hole size, but is used in this model to demonstrate the difference between the two types of stabilisers.

The bottom stabiliser is a more usual spiral bladed stabiliser, with 12.125" OD blades and a 10 ft long 9.5" body.



Fig. 31

Two different bit models are available, a PDC model and a blank tri-cone body. These are used to illustrate the relative importance of bit flow by area between bit types. In this case, however, only the PDC bit is used. The OD of the stabilisers is painted solid red or black, and the body area is hatched in red to provide clear indication of the various components once inside the tube.

The cuttings are modelled using fine grain table salt and course sea salt. These simulate cuttings and fines of scale dimensions.

The picture below shows the cross section of the PDC bit model, clearly illustrating the flow by area.

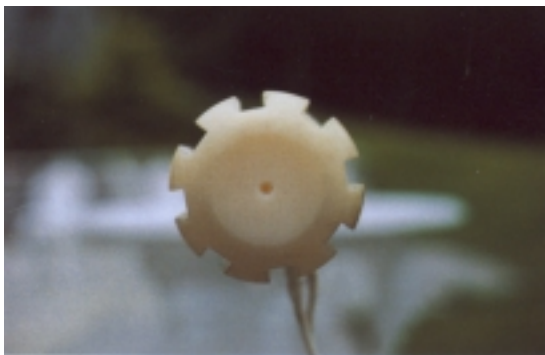


Fig.32

The BHA is guided into the Perspex tube while the assembly is pulled using the rope attached to the top of the drill pipe.

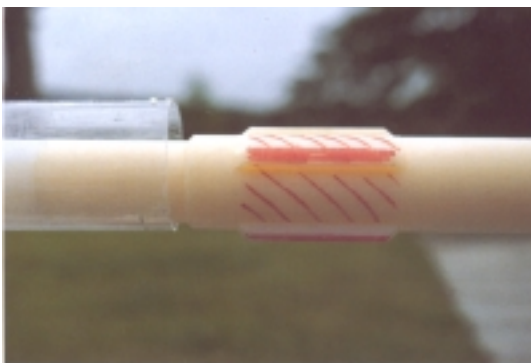


Fig.33

The lower section of the BHA can be seen prior to entering the tube. The stabiliser and bit are clearly visible.



Fig.34

The drillpipe/Collar crossover is shown ‘shovelling’ a substantial pile of cuttings ahead of the change in cross sectional area. This is a scale distance of 40 ft above the top stabiliser.

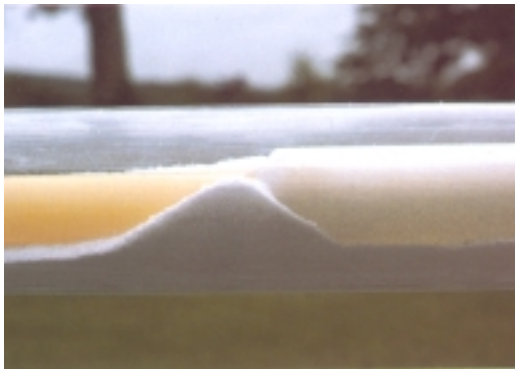


Fig.35

ABC of Hole Cleaning

After pulling the BHA a further scale distance of 6ft into the model, a pile of cuttings ahead of the Drillpipe/Collar can be seen to increase in height.

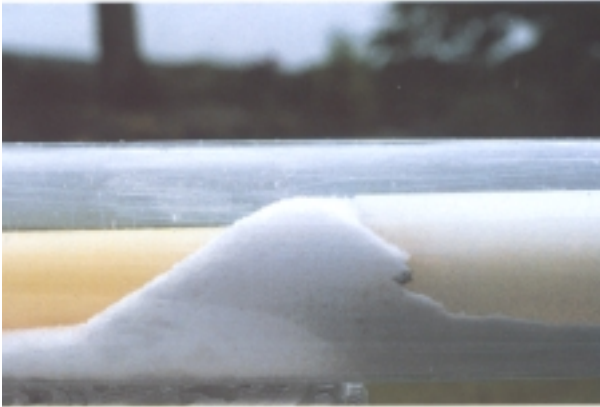


Fig. 36

The top stabiliser enters the tube and cuttings begin to build up.

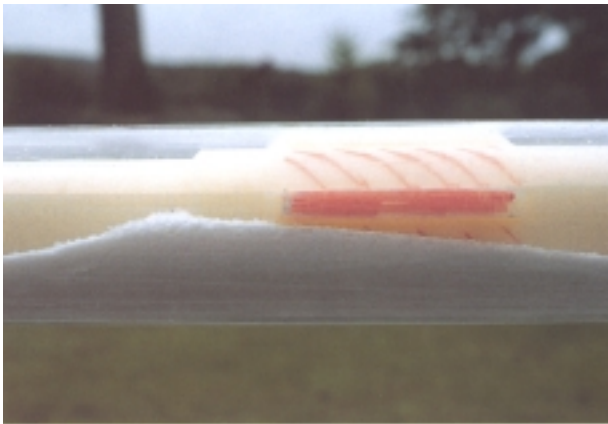


Fig.37

As the BHA is drawn further into the tube, the cuttings can be seen to build up around the stabilisers.



Fig.38

The straight bladed stabiliser has less of a shovelling effect than the spiral stabiliser. The difference in thickness of the cuttings bed after the BHA has passed can be seen in the picture below. Below the stabilisers (i.e. to the right) very few cuttings remain on the low side of the tube.



Fig.39

As the BHA is drawn further through the tube, a significant pile of cuttings builds up in front of both stabilisers. Again, the bigger pile is in front of the spiral stabiliser.

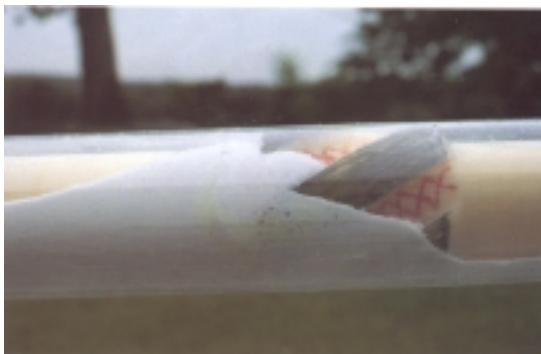


Fig. 40

The gap at the top of the annulus had now closed and the stabiliser is effectively packed off with cuttings. The overpulls now increase rapidly and the string will become stuck in a short time.



Fig.41

Here, an overview of the two stabilisers and the cuttings forming around them can be seen clearly.



Fig.42

The picture shows how the cuttings are dragged ahead of the stabilisers, leaving very few behind to cause problems at the bit. If the flow-by area of the stabiliser were not as restrictive, then the piling of the cuttings would occur at the bit. Due to the lower flow-by area of the bit, the piling up of cuttings would occur over a shorter distance.

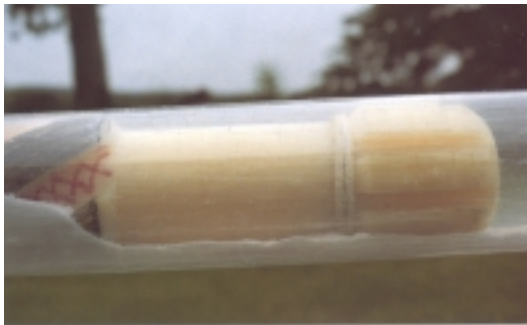


Fig.43

Summary of the Hole Cleaning Model

1. The model illustrates how the cuttings can build up in front of stabilisers and other changes in cross sectional area.
2. It can be seen from the model why jarring up when getting stuck while pulling out of the hole can be the wrong thing to do.
3. The model is aimed at situations where gauge or close to gauge hole exists. Over gauge hole will give fewer problems with cuttings build up as the flow-by area around the BHA components will effectively be greater.
4. The depth of a cuttings bed that will cause problems while pulling out of hole is surprisingly small.
5. The use of the model in the classroom situation of the Stuckpipe Prevention Course was a significant benefit to the learning of the attendees. Many comments were received on how well this model enabled offshore staff to visualise what was happening downhole.

7. Solids Removal at surface

The objective of this section is to describe the function and operation of surface solids removal equipment.

The surface solids removal equipment is designed to remove the unwanted solids from the drilling fluid, while maintaining the maximum amount of fluid. A well-designed solids removal package will provide significant benefits if it is able to cope with all the possible conditions when drilling. There are several factors to consider when selecting a solids removal system:

- a. mud type (obm/wbm)
- b. Hole size (smaller hole sizes give less and smaller cuttings)
- c. Bit type (an aggressive bit makes a bigger cutting)
- d. Hole Type (vertical/deviated)
- e. Pump rate (limits on pump pressure)
- f. Lithology (reactive clays to be drilled)
- g. Logistics (enough supply of drilling fluids/chemicals)

The environmental restrictions on discharges of drilling fluids may also have a large impact on the selection of equipment.

The primary solids removal devices in order of operation are:

- a. Shale Shaker
- b. Sand Trap (settling tank)
- c. Hydrocyclone desander
- d. Hydrocyclone desilter
- e. Centrifuges

Solids are generally divided into two classes:

- a. High specific gravity solids, s.g. 4.2 kg/l (barites) – these are generally not removed by the shaker screens.
- b. Low specific gravity solids, formed from the drilled cuttings – these have an average density of 2.6 kg/l

The particle size also has an influence on the drilling fluid properties (viscosity), with the smaller particles having a greater effect on the fluid.

Particle Size (microns)	Particle Classification
>2000	Coarse
2000-250	Intermediate
200-74	Medium
74-44	Fine
44-2	Ultra-fine
<2	Colloidal

Shale Shaker

The shale shaker is the first device in the solids removal chain. The finer the screen that is on the shaker, the more solids that are removed at this early stage.

Over the last decade, improvements in shale technology have led to the use of finer screens. The particle size removed by the shaker depends entirely on the size and shape of the openings in the screen cloth. On a shaker with several screens in a series arrangement, the finest mesh screen determines the particle size.

As drilling fluids become more complex and the environmental constraints increase, the trend toward finer shaker screens will continue. It is important, therefore, that they are run as effectively as possible. Their efficiency depends on the number of units available, their maintenance condition and the suitability of the screens installed.

Cuttings monitoring

The volume of cuttings at the shakers should be closely monitored and an estimate made to compare this with the theoretical volume cut by the bit.

Screen and particle sizes

The table and illustration below provides an overview of the various screen sizes. Square mesh and oblong screens are available for most shakers.

Below are given the VSM 100 Hook Strip Screens

Mesh	Opening Width
8 x 8	2465 μ
10 x 10	1976 μ
20 x 20	895 μ
30 x 30	567 μ

Primary screens for the VSM 100

Mesh	Opening Width
52 x 52	340 μ
84 x 84	215 μ
105 x 105	165 μ
120 x 120	150 μ
145 x 145	120 μ
165 x 165	105 μ
200 x 200	87 μ
230 x 230	75 μ
300 x 300 *	49 μ
325 x 325 *	42 μ

(The majority of particles size of barytes ranges from 45 μ - 75 μ)

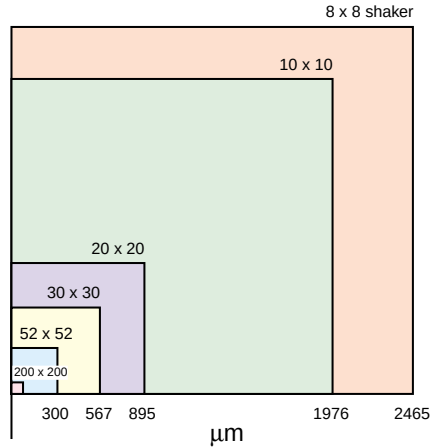


Figure Shaker screen sizes:
40x Enlargement of screen mesh sizes
from 200 x 200 to 8 x 8.

The liquid capacity of the shakers depends primarily on the removal of solids from the screens, assuming that the mud properties and formation solids remain equal.



Fig.46 - Shale Shaker

This capacity will be reduced if the openings in the screen become clogged with sand or other particles, or if the screen becomes coated with sticky un-

dersized particles in a mud. If a screen becomes blinded, there are three options:

- Change the screens to a coarser mesh if the problem is undersized solids, or a finer mesh if the problem is near size plugging.
- Allow the situation to continue. This will cause mud to be lost at the shaker, wasting money and cause an undesirable environmental situation. Not an option for environmentally sensitive drilling fluids.
- Bypass the shaker, allowing all the solids to pass downstream, with the possible chance of plugging equipment or drillpipe.

The more severe the conditions, the more important the design and operation of the shaker becomes. Evidence has shown that frequent cleaning of the screen reduces the oil on cuttings discharge figures. Secondary drying screens can reduce fluid discharges further.

Since the shaker is the most important item in the solids control chain, it is necessary to continually monitor it, checking for screen tears, etc. The quality and quantity of cuttings should also be monitored to allow the cuttings removal techniques to be optimised.

Inefficient operation results in lower throughput and higher screen consumption.

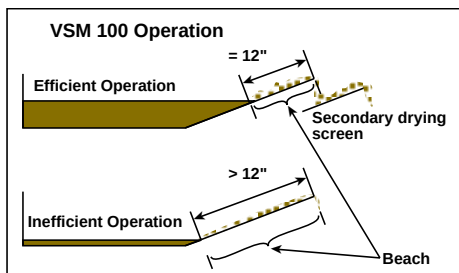


Fig.47 - Operation of the VSM 100 shale shaker

Sand Trap

The sand trap catches all the particles that go through the shaker screens, and, as such, is an important part of the solids removal chain. The trap receives all the fluid that is discharged from the shakers and acts as a settling tank, allowing all the particles that pass through the shaker to settle. The sand trap should never be bypassed when drilling.

The sand trap will also catch any larger particles that have passed through the shaker if it has damaged screens, etc.

The traps should be dumped frequently when using a water based mud and drilling in an area with a high sand content. Dumping of oil-based mud is not allowed, therefore the sand and cuttings in the sand traps must be processed by downstream solids removal equipment.

Desander

The desanders are 6" – 12" ID hydrocyclones and have a cut of about 50 microns. They should be setup to process 1.5 times the maximum pump output.

Fluid is pumped into the desanders tangentially near the top of the cone, causing it to spiral downwards. The solids are then pushed out by the centrifugal force, and move downwards under the influence of gravity, to be discharged at the bottom. The clean drilling fluid comes out of the top.

The operation of any hydrocyclone needs to be carefully monitored as it will discharge large amounts of fluid if not operated efficiently.

Desilter

A desilter is a smaller size of hydrocyclone, around 4" in diameter. They operate in a similar way to desanders, and remove particles in the range of 20 – 40 microns.

Centrifuges

The centrifuge increases the rate of settling of solids by the use of centrifugal force. In the North Sea, a minimum of two centrifuges is recommended.

They can be operated in two ways:

- Solids removal mode
- Barytes recovery mode

When in Barytes recovery mode, particles larger than 4 – 7 microns are removed. In solids removal mode, particles bigger than 6 – 10 microns are removed. The main use of the centrifuge is to control viscosity, by removing the collidal part of the fluid. These are very fine particles, generally <2µm.

8. Well planners guide to hole cleaning

The objective of the section is to highlight the importance of drillpipe size, and the effect that changing the size will have on the annular velocity of the drilling fluid.

It is important that well planners know the implications of their design on hole cleaning. There are several key areas which impact this:

- Trajectory
- Mud Rheology
- Surface Equipment
- String Design
- Time Allocation (especially circulating hole clean times)
- Procedures
- Correct mud gradient
- Casing setting depths

Trajectory

The trajectory of the well may be dictated by other wells in the area, the target and economics. It is important to consider the implications of certain trajectories:

For example, 17.5" holes with inclinations $>55^\circ$ are best avoided, if possible.

Surface Equipment

This includes flow lines, pumps and shakers. The equipment is essential for good hole cleaning.

Drill String

The selection of the main drillstring components for pressure loss and flow rate is crucial.

A trend seen in 17.5" hole section in the past is for maximum flow rate to be sacrificed at the expense of extra equipment in the drillstring, e.g. LWD, MWD, motors, etc. 17.5" hole sections should be designed to allow 1000 – 1100 GPM to be used to TD after taking into account all the equipment pressure losses.

For a given hole size and drillpipe size, there will be a range of annular velocities that can be obtained with the pumps and lines available on the rig. The main limitation is the pressure available from the pumps. If the flow rate is increased for the same pressure, then the annular velocity would increase.

The most convenient way of doing this is to reduce the restrictions in the well bore.

The main way to reduce the pressure losses is by hydraulics optimisation and the removal of any downhole equipment which may cause large pressure drops. For example, increasing the OD of the drillpipe will have two effects:

1. The pressure drop in the drillstring will be reduced and allow higher pump speeds for the same pressure.
2. The decrease in size of the annulus will force an increase in the annular velocity by a small amount (about 6% when going from 5" to 6-5/8" drillpipe in 17-1/2" hole).

The ability to increase the annular velocity by using a larger drillpipe is mainly due to the reduced pressure drop inside the drillpipe. Increase in the annular velocity due to the larger OD is a lesser effect.

The following graphs illustrate the effect of various drillpipe sizes. Graphs 48a, b & c show that simply increasing the drillpipe size without increasing the flowrate will not substantially increase the Annular Velocity.

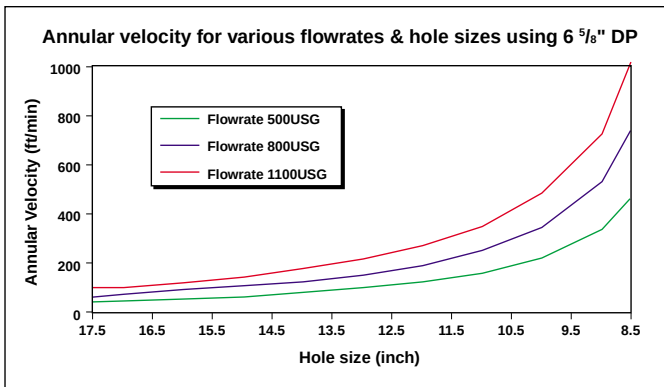


Fig.48a

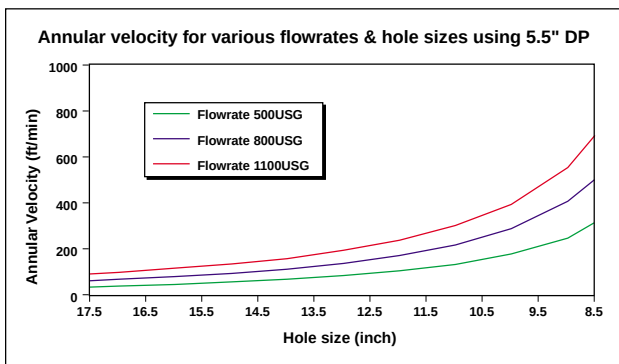


Fig. 48b

The flowrate at which the drilling fluid becomes turbulent inside the drillpipe is approximately:

5" – 75 GPM

5.5" – 85 GPM

6-5/8" – 98 GPM

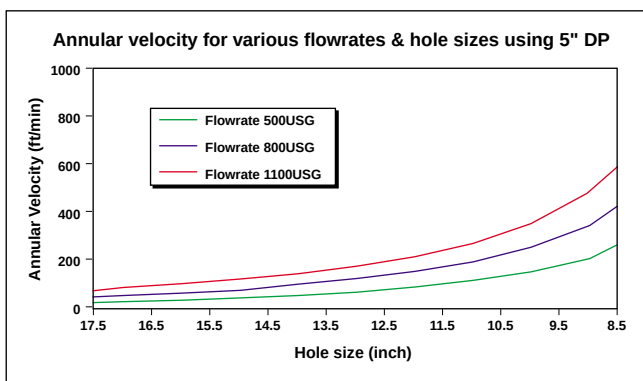


Fig. 48c

Graph 49 illustrates how an increased drillpipe size gives less pressure drop and allows an increased flowrate, providing a significant increase in annular velocity.

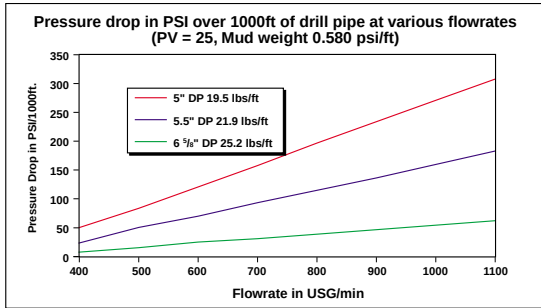


Fig. 49

Graph 50 illustrates the advantage of larger drillpipe by comparing stand pipe pressure for an example well. Two hole sizes are illustrated, with three sizes of drillpipe for each hole size.

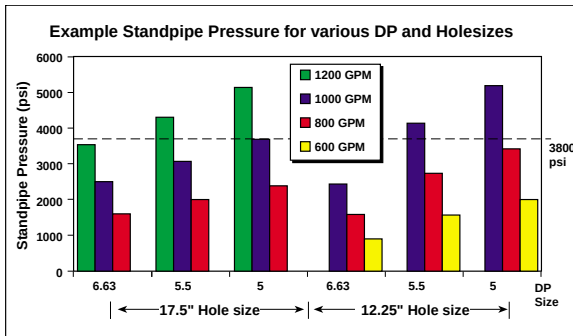


Fig. 50

For graph 50 the maximum pump output attainable without exceeding the allowable surface pressure is taken and used to look up the annular velocity in graphs 48a – c.

For a 17.5" hole, a flow rate of 1200 GPM is only attainable using 6-5/8" drillpipe, giving an annular velocity of 120 ft/min. A flow rate of 1000 GPM is only attainable using 5.5" drillpipe, giving an annular velocity of 90 ft/min. Using 5" drillpipe limits the flow rate to 800 GPM, with an AV of 65 ft/min. It is interesting to note that all of these AV values are too low for effective hole cleaning.

For a 12.25" hole, a flow rate of 1000 GPM is only attainable using 6-5/8" drillpipe, giving an annular velocity of 230 ft/min. A flow rate of 800 GPM is only attainable using 5" drillpipe, giving an annular velocity of 165 ft/min.

The graph below shows the same for a 16" hole, assuming the maximum pump pressure available is 3800 psi.

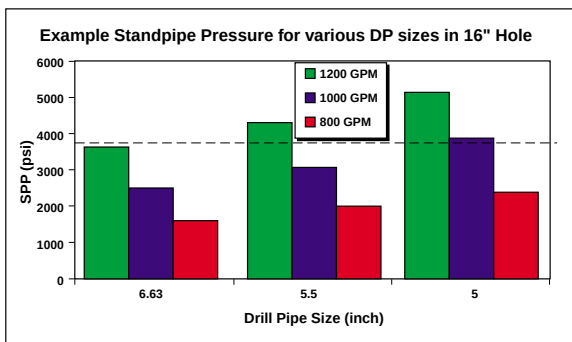


Fig. 51

For a 16" hole, a flow rate of 1200 GPM is only attainable using 6-5/8" drillpipe, giving an annular velocity of 160 ft/min. A flow rate of 1000 GPM is only attainable using 5.5" drillpipe, giving an annular velocity of 160 ft/min. Using 5" drillpipe limits the available flowrate to 900 GPM, giving an AV of 100 ft/min.

9. Notes

General

1. Think downhole – where are the problems; how can I drill faster
2. Plan trips with a picture of the well in mind – where are the cutting likely to be
3. Know how to interpret the signs
4. Know how to react to the signs
5. React by: designing out problems
 - i.e. Tandem Pills
 - 6-7/8" pipe
 - Good Shakers

Cleaning the hole before pulling out

1. In deviated wells (45° plus), the hole volume needs to be circulated at least twice to ensure cutting & caving reach the surface.
2. To save some time, circulate the open hole volume at least twice before pulling into the shoe. When in the shoe, pump as fast as possible to lift the ratings while rotating – we are not worried about hole erosion at this point (carving wear may be a problem, though).
3. Insert circulating subs into the BHA to ensure maximum pumping rates if a restriction is imposed by downhole motions, etc., while drilling. The sub can then be opened when in the casing.
4. Keep cleaning pills to a minimum by running an adequate lower end rheology at all times when drilling.

Possible signs of poor cleaning

1. Excessive torque and drag
2. Tight connections
3. Pumping out of the hole
4. Hole packing off on connections
5. Decrease in cuttings volume over the shakers
6. Rapid increase in MBT due to grinding shale particles
7. Increase in PV with no chemicals being added.

Evidence of any of the above indicates that the hole is not being cleaned effectively.